Philipp Sepin

psepin.com GitHub LinkedIn ORCID

Education

TU Wien, *MSc Computational Science and Engineering* Feb 2024 – Feb 2027

- High-performance computing, GPU and multiprocessor programming.
- Machine learning and computational chemistry.
- Master's thesis on interpretability of graph foundation models.

TU Wien, *Additional Minor in Computer Science* Feb 2023 – Feb 2025

- Additional 30-ECTS program, done alongside BSc and MSc studies.
- Introductions to algorithms, data structures, databases and security.

TU Wien, *BSc Technical Physics* Oct 2019 – Feb 2024

- Physics and mathematics.
- Electives focused on machine learning and statistics.

HTL Lastenstrasse 1, *Higher Technical School* Sep 2013 – Jun 2018

- Electrical engineering and electronics.
- Honors class / program.

Peer-Reviewed Publications

Mapping Mining Areas in the Tropics from 2016 to 2024 2025
Sepin, P., Vashold, L., and Kuschnig, N.
Nature Sustainability.
DOI: 10.1038/s41893-025-01668-9
Interactive Map Repo

Comparison of Clustering Algorithms for Statistical Features of Vibration Data Sets 2023
Sepin, P., Kemnitz, J., Rezapour Lakani, S., and Schall, D.
Proceedings of the 5th International Data Science Conference.
DOI: 10.1007/978-3-031-42171-6_1

Experience

WU Vienna University of Economics and Business, *Research Assistant* Vienna, Austria
Aug 2023 – Jan 2025

- Developed and trained segmentation models for high-resolution satellite imagery across multiple GPUs (87.7% accuracy). Built end-to-end pipelines processing 23 TB of geospatial data on an HPC cluster.
- First author of the resulting *Nature Sustainability* paper, introducing the first panel dataset of mining areas across the tropics (147,000+ polygons, 2016 to 2024). Received the WU Award for a Star Journal Article.

Siemens, *Student Researcher* Vienna, Austria
Aug 2022 – Feb 2023

- Designed pipelines for semi-supervised learning on time-series vibration data as part of my bachelor's thesis in the Distributed AI Systems Research Group.
- Published and presented at a local data science conference.

Philips, *Intern* Klagenfurt, Austria
Jun 2019 – Aug 2019,
Jul 2020 – Sep 2020,
Jul 2021 – Sep 2021

- Built and optimized real-time image classification models for embedded systems.
- Trained image and text classification models for spam filters on AWS EC2.

Hatec Automatisation, *Working Student* Klagenfurt, Austria
Aug 2017 – Apr 2018

- As my final year project in the higher technical school, I implemented an automated sorting system with two robots, conveyors and object recognition.

Other Selected Projects

NanoGPTQ 2025

- Implemented a small GPT model and GPTQ int8 quantization from scratch.
- Reduced model size 4× (fp32 to int8) with no loss in perplexity. Repo

TTA for Molecular Solubility Prediction 2025

- Explored test-time adaptation for graph neural networks in molecular property prediction. Repo

Skills

Proficient: Python
Familiar: PyTorch, C++, CUDA, R, MPI, OpenMP, SQL, GIS
Tools: Git, Bash, Slurm, LaTeX
Interests: Ski mountaineering, climbing, analog photography

References

Dr. Esther Heid, who supervised me during the molecular test-time adaptation project.
Assistant Professor, TU Wien
hesther.github.io

Dr. Nikolas Kuschnig, who supervised me during the mining research project.
Senior Lecturer, Monash University
kuschnig.eu